Tyler Roberts

Senior Software Engineer

CONTACT

tyler.roberts.1991@outlook.com
+1 6142073008
Florida, United States
LinkedIn: linkedin.com/in/tyler-roberts-a7b0953a0

SKILLS

Programming
Python, JavaScript, TypeScript, Node.js

Data & ML
Agentic AI, LLM Prompt Engineering, DSPy, Retrieval-Augmented Generation (RAG), Evaluation Pipelines, Anomaly Detection

Data Engineering
Dagster, Airflow, ETL Pipelines, Backfills, Data Quality Monitoring

Cloud
AWS, ECS, Lambda, EventBridge, S3, Glue

Backend & APIs
REST APIs, Distributed Systems, Microservices, Metadata Services

DevOps & Observability
CI/CD, Docker, Structured Logging, Metrics, Monitoring & Alerting

Integration
Git & PR Metadata, Code Diffs, Branch & Merge Tracking, Third-Party APIs

EDUCATION

Florida State University
Bachelor of Computer Science
2009 - 2013

Professional Summary

Senior Software Engineer with 9+ years of experience designing and delivering production-grade data, ML, and cloud-native systems in large-scale distributed environments. Strong expertise in Python-based data pipelines, agentic retrieval workflows, and LLM-driven scoring systems that operate on PR, commit, and code-quality metadata at scale. Proven track record building orchestration-heavy platforms using Dagster and AWS services with a focus on reliability, observability, and cost-aware execution. Hands-on experience optimizing LLM prompting strategies, evaluation loops, and feedback-driven corrections using modern prompt optimization techniques including DSPy-style workflows. Adept at collaborating with platform, data, and product teams to translate ambiguous productivity and quality metrics into scalable, automated systems that improve engineering outcomes.

Professional Skills

Senior Software Engineer

Web Venturez LLC | February 2020 - October 2025

- Designed and implemented large-scale backend systems that ingest, normalize, and analyze pull request, commit, and branch metadata to generate engineering quality and productivity signals.
- Built Python-based data pipelines and orchestration workflows using Dagster to support scheduled processing, historical backfills, retries, and data quality checks across AWS infrastructure.
- Developed agentic retrieval workflows that dynamically assemble code context, diffs, and historical signals to improve LLM-driven scoring accuracy and reduce hallucinations.
- Led prompt optimization efforts for LLM-powered scoring and classification systems, balancing cost, latency, and quality through iterative evaluation, feedback loops, and prompt tuning strategies.
- Implemented evaluation pipelines and feedback-driven correction mechanisms to continuously refine CES-style scoring outputs based on reviewer and user input.
- Integrated backend services with APIs and internal dashboards to expose PR-level metrics, historical trends, and anomaly detection insights to engineering stakeholders.

Full Stack Engineer

Webiz Digital | July 2017 - January 2020

- Developed backend services and data processing workflows supporting code-quality analytics, reporting dashboards, and internal tooling used by distributed engineering teams.

- Built and maintained Python and Node.js services that consumed Git-based metadata, linked commits to pull requests, and tracked merges across multiple repositories.
- Collaborated with product and platform teams to design data models for effort, quality, and work-type classification aligned with business and engineering metrics.
- Implemented reliable ETL-style pipelines with retry logic and error handling to ensure consistency across high-volume data ingestion processes.
- Contributed to early experimentation with ML-assisted classification and scoring logic to improve engineering visibility and reduce manual review effort.
- Supported production systems through monitoring, debugging, and performance optimization in cloud-hosted environments.

Software Developer

Centric Tech Inc. | October 2013 - June 2017

- Built and maintained backend applications and data-processing components that supported internal analytics and operational workflows.
- Implemented APIs and services that aggregated structured and semi-structured data from multiple sources for reporting and analysis.
- Worked closely with senior engineers to design scalable data models and processing logic suitable for growing datasets and evolving requirements.
- Assisted with reliability improvements, including logging, monitoring, and error handling for production systems.
- Participated in debugging, refactoring, and performance tuning efforts to improve system stability and throughput.
- Established a strong foundation in software engineering best practices that later enabled ownership of complex, production-grade systems.